

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **production ML pipelines** processing **billions of rows daily** across **PostgreSQL, MySQL, Oracle** databases; automated data preparation, model training, evaluation, and deployment cycles reducing manual intervention by **70%** and enabling real-time inference at scale.
- Developed **LLM-based solutions** using **Claude API (Anthropic)** with advanced **prompt engineering, RAG architectures**, and structured-JSON constraint guardrails; orchestrated end-to-end **generative AI workflows** delivering tailored content generation at scale with **95%+ accuracy** in production.
- Built **automated ML infrastructure** using **Python, pandas, NumPy**, and **scikit-learn** for predictive analytics, anomaly detection, and time-series forecasting; integrated **distributed data systems** with **AWS Lambda** and **CloudWatch** to monitor model drift and trigger retraining workflows, reducing false positives by **45%**.
- Delivered **enterprise data platform integrations** leveraging **REST APIs** and **SQL-based data flows** across interconnected systems; applied **BI concepts** and **data modelling** techniques to design schemas supporting **petabyte-scale** analytics workloads with **sub-second** query latency.
- Engineered **production-grade ML code** with **CI/CD automation** using **Jenkins, Git**, and **GitHub Actions**; containerized ML services with **Docker**, deployed to **AWS EC2**, and established comprehensive monitoring and alerting for model performance and data quality across enterprise business functions.
- Collaborated with cross-functional stakeholders—business analysts, operations teams, and domain experts—to scope, validate, and ship **AI and ML systems** solving real-world enterprise problems; conducted UAT support and post-implementation optimization, achieving **SLA compliance of 99.5%+** in production environments.

SKILLS

Machine Learning & AI – ML model development, LLM-based solutions, generative AI systems, prompt engineering, RAG architectures, agentic frameworks, anomaly detection, predictive analytics, time-series forecasting, supervised/unsupervised learning, model evaluation metrics.

Programming & Data Engineering – Python, SQL (PostgreSQL, MySQL, Oracle), data wrangling, ETL pipeline development, automated data preparation, distributed data processing, data warehousing, feature engineering.

Production ML Systems – ML pipeline automation, model training/evaluation/deployment, model drift monitoring, inference serving, production code quality, debugging pipelines, real-world feedback iteration.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, CI/CD (Jenkins, Git, GitHub Actions), Linux, shell scripting, log analysis.

Data Science & Analytics – BI concepts, Power BI (dashboards, DAX, Power Query), Tableau, data modelling, exploratory data analysis, statistical analysis, A/B testing, R/RStudio.

Enterprise Platforms & Integration – REST API integration, enterprise data platforms, system interoperability, change-control management, cross-system data flows, Jira/Confluence.

Specialized Domains – Claude API (Anthropic SDK), pandas, NumPy, scikit-learn, Flask, FastAPI, prompt engineering, workflow orchestration.

EDUCATION AND TRAINING

PROJECTS

Ireland Jobs Tracker – Generative AI Resume Automation Platform (Personal Project)

- Architected end-to-end **generative AI workflow** integrating **Claude API (Anthropic)** with **prompt engineering** techniques to generate job-description-tailored resumes at scale using **claude-haiku-4-5** and structured-JSON constraint guardrails.
- Orchestrated multi-stage **ML pipeline**: web scraping via **Apify** for JD ingestion, **LLM-based content generation**, **Python-based PDF rendering** with reportlab, and continuous deployment via **GitHub Actions** to **Vercel**; automated end-to-end processing of **100+ job descriptions monthly**.
- Applied advanced **prompt engineering** and **agentic design patterns** to constrain LLM outputs, ensuring consistency and business-rule compliance; demonstrated hands-on expertise in **LLM integration**, **API orchestration**, and **production-grade AI system development**.

Enterprise SWIFT Payments Operations & Monitoring Dashboard (HCL Technologies)

- Designed and deployed **production ML and data systems** for real-time transaction monitoring, anomaly detection, and SLA-critical alerting across **SWIFT messaging** (MT/MX formats) and **ISO 20022 message flows**; processed **billions of payment records daily** with **99.9%+ uptime** and **sub-100ms** query latency.
- Built **automated data pipelines** using **Python ETL**, **SQL** (PostgreSQL, Oracle), and **distributed data systems** to ingest, validate, and transform payment transaction data; applied **anomaly detection algorithms** and **time-series forecasting** to identify operational anomalies, reducing false negatives by **60%**.
- Integrated **ML monitoring and alerting infrastructure** using **Splunk**, **Dynatrace**, **CloudWatch** dashboards with **BI visualization**; partnered with banking operations and business stakeholders to scope data requirements, validate model outputs, and ensure **regulatory compliance** (GDPR, audit trails, payments-processing standards).

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)